

Primary Key

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@Id

- **Purpose:** Marks a field as the primary key of the entity. Every entity must have a primary key.
- **Usage:** Placed on a field (or getter method).
- **Example:**
Java

```
@Id  
private Long id;
```

@GeneratedValue

- **Purpose:** Specifies that the primary key value will be generated automatically by the database.
- **Attributes:**
 - **strategy:** Defines the generation strategy. Common strategies include:
 - `GenerationType.AUTO`: (Default) Lets the persistence provider choose the best strategy for the underlying database.
 - `GenerationType.IDENTITY`: Uses a database identity column (e.g., `AUTO_INCREMENT` in MySQL, `SERIAL` in PostgreSQL).
 - `GenerationType.SEQUENCE`: Uses a database sequence. Requires `@SequenceGenerator`.
 - `GenerationType.TABLE`: Uses a separate table to simulate a sequence. Requires `@TableGenerator`.
 - **generator:** Name of the `SequenceGenerator` or `TableGenerator` to use.
- **Example (Identity):**

```
@Id  
@GeneratedValue(strategy = GenerationType.IDENTITY)  
private Long id;
```

- **Example (Sequence):**

```
@Id  
@GeneratedValue(strategy = GenerationType.SEQUENCE, generator = "my_seq_gen")  
@SequenceGenerator(name = "my_seq_gen", sequenceName = "my_sequence", allocationSize = 1)  
private Long id;
```

